Ref: Introduction to mechatronic design, J. Edward Carryer, R. Matthew Ohline, Thomas W. Kenny

switch could be purchased for less than 25 cents in high quantities (1,000s or more). When low cost and simplicity are strict requirements, reed switches are an excellent choice.

Reed switches are used in a very large number of applications, including automobiles, home appliances, and toys. One familiar example is a bicycle computer, which senses wheel rotation and sometimes pedal rotation with reed switches. A small magnet is mounted to the spokes of a bicycle's front wheel, and a reed switch is mounted on the front fork so that the magnet passes very near the switch as the wheel rotates. Each time the magnet passes the reed switch, the contacts close and the cycle computer calculates quantities such as speed and distance.

13.5.5 Proximity Sensors

Proximity detection is usually performed in the digital domain with a sensor that detects when an object is within the range of the sensor. Proximity sensors give a "yes or no" answer to the question of whether an object is present or absent. Usually, the range of these sensors is small: rarely more than a few centimeters. A common application of proximity sensing is when a motorized stage in a piece of automated equipment reaches the home, or zero, position. Several of the sensors we've already described above are used for proximity detection, and are commonly used to detect the home position for motion stages. These include contact switches, microswitches, opto-interrupter emitter–detector pairs, reflective emitter–detector pairs, Hall switches, magnetoresistive switches, and reed switches. To round out the discussion, we present three other commonly used proximity sensors.

13.5.5.1 Capacitive Proximity Sensors

Capacitive proximity sensors detect changes in the capacitance between two plates of a capacitor when an object moves close enough to measurably disturb the electric field. An example of a **capacitive proximity detector** is the Omron E2K-X, which is shown in Figure 13.55. This noncontact sensor can detect a wide variety of metallic and nonmetallic objects like glass, wood, water, and plastic at distances ranging from 3 to 30 mm. These characteristics make it a good choice for use in automated production lines.



FIGURE 13.55

A capacitive proximity sensor (Omron E2K-X). (Courtesy of Omron Electronics, LLC.)

Capacitive proximity sensors have also become very popular in user interfaces. Capacitive touch sensors capable of detecting the presence of one or more fingers are now commonplace in a wide variety of applications, such as computer touch pads, mp3/media player controls, and mobile phone touch screens.

13.5.5.2 Inductive Proximity Sensors

Inductive proximity detectors use an approach similar to that of capacitive proximity detectors, except that they use an oscillating signal in a coil to establish an electromagnetic field rather than an electric field. Inductive sensors detect when a metallic object has moved close enough to measurably disturb the magnetic field. Nonmetallic objects do not disturb the

field, and are not detected by inductive sensors. Figure 13.56 shows a typical inductive sensor, the Omron E2F, which closely resembles a capacitive proximity sensor. This sensor can detect metallic objects that are up to 8 mm away.



FIGURE 13.56

An inductive proximity sensor (Omron E2F). (Courtesy of Omron Electronics, LLC.)

13.5.5.3 Ultrasonic Proximity Sensors

Ultrasonic proximity detectors emit high-frequency sound waves and measure the amount of time required for reflections to return. Shorter return times indicate that an object has moved into close proximity. Figure 13.57 shows the Omron E4C-DS100 ultrasonic proximity sensor, which has a range of 9 to 100 cm. Any object that reflects at least a portion of the ultrasonic pulse will be detected, regardless of the material. Note that this sensor also closely resembles the capacitive and inductive sensors discussed above, but that the piezo element at the distal tip hints that this is an ultrasonic sensor.



FIGURE 13.57

An ultrasonic proximity sensor (Omron E4C-DS100). (Courtesy of Omron Electronics, LLC.)

13.5.6 Position Sensors

Distinct from the task of reporting whether or not an object is within range of a sensor is the task of determining the distance from a sensor to a location or the distance that something has moved. These are the jobs of **position sensors**, also called **displacement sensors**. In this section, we present a selection of the most commonly used linear and angular position sensors.

13.5.6.1 Potentiometers

One of the simplest position sensors is the humble potentiometer. A **potentiometer**, often referred to in the diminutive as a **pot**, is a three-terminal device with a fixed resistance between two terminals, and the third terminal is connected to a movable **wiper** that contacts the resistance somewhere midspan. This creates two adjustable resistances that depend on the relative position of the wiper between the two ends of the resistance. The position of the wiper may be determined by measuring either of the variable resistances. The schematic symbol for a potentiometer is particularly effective at illustrating its construction and function (Figure 13.58).

Recall from the discussion in Section 13.3.2 that we chose a potentiometer as a simple means of determining the displacement of a mass on a spring to infer acceleration. In general,

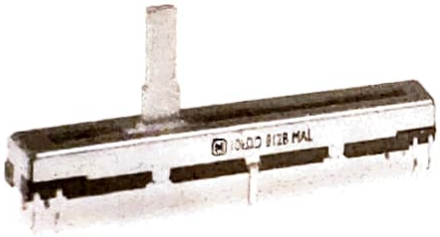


FIGURE 13.58
Schematic symbol for a potentiometer.

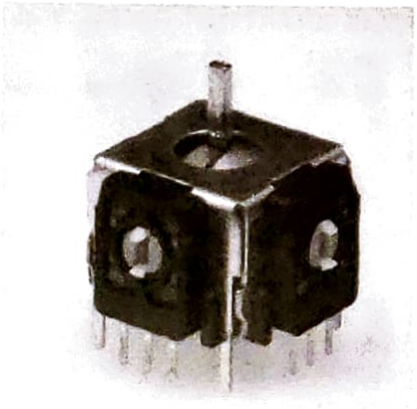
potentiometers are not particularly high-performance position measurement sensors, as they add frictional loads to the system being measured, and are subject to temperature effects, vibration, and long-term drift resulting from wear and tear. However, they are very inexpensive and readily available in a wide variety of packaging options and resistance values. They are often the first choice to consider for position measurements where requirements aren't highly demanding. Potentiometers are currently used to determine position in a great many applications. For example, rotary and linear pots (Figure 13.59a and b) are used as user input devices on stereo equipment. Potentiometer-based joysticks are used in video game consoles user interfaces (Figure 13.59c). Another interesting and useful variant is the **cable-driven linear displacement transducer** (more commonly called a **string pot**), which is comprised of a precision multiturn rotary potentiometer whose input shaft is attached to a constant tension torsional spring and a spool holding a long string. This device shown in Figure 13.59d is made by UniMeasure, Inc. The resistance of the string pot output is proportional to the length of string that has been pulled from the spool. This is a simple and effective means of measuring relatively large distances (up to about 3 m/10 ft is practical) at low cost with impressive accuracy (around 0.5%), repeatability (0.05%), and linearity (0.5%).



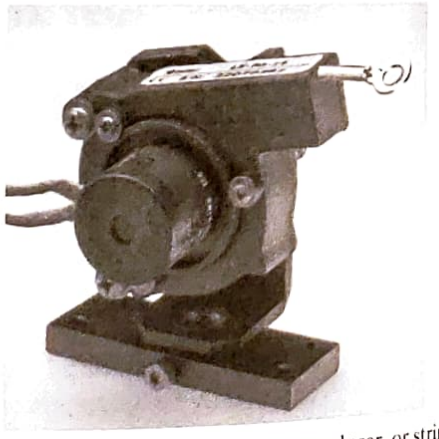
(a) Rotary potentiometer



(b) Linear potentiometer, or slide pot



(c) Joystick



(d) Cable-driven linear displacement transducer, or string pot

FIGURE 13.59
A selection of potentiometers.

When used to measure position, potentiometers are in effect **absolute position sensors**: the output corresponds to the location of the wiper and does not require a separate position measurement to be used as a relative reference or serve as the origin. This contrasts with **relative position sensors**, which measure incremental motion away from an initial position.

13.5.6.2 Optical Encoders

A very popular type of position sensor that is available in both relative and absolute versions is the **optical encoder**. The most common approach used in commercially available devices is illustrated in Figure 13.60. Light from an LED is directed toward an optically opaque material such as a thin stainless steel disk that has a series of holes in it. When the disk rotates relative to the light source, a photo-diode on the opposite side detects transitions in the levels of light that occur as the light is alternately blocked or allowed to pass through a hole. Counting the transitions enables the user to track any rotation that occurs. Because of the incremental nature of these measurements, these sensors are often called **incremental encoders**. Very high resolutions are possible with this approach, as the holes can be made to be very tiny and closely packed through the use of manufacturing techniques such as chemical etching or laser cutting. Resolutions of better than 500 counts per turn are readily available.

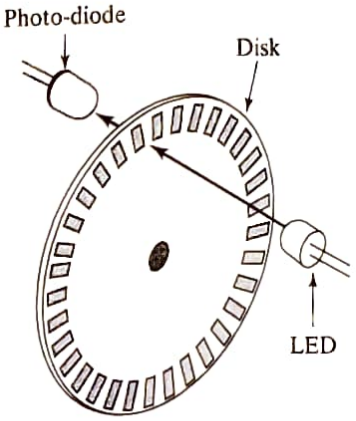


FIGURE 13.60
Elements of a rotary optical encoder.

Quadrature encoding can be used to determine the direction of motion or rotation. By using two photo-diodes (labeled A and B in Figure 13.61) to detect light transitions and off-setting them by half the spacing between holes so that one sensor is blocked when the other is illuminated, the relative phase of the resulting signals indicates the direction of motion.

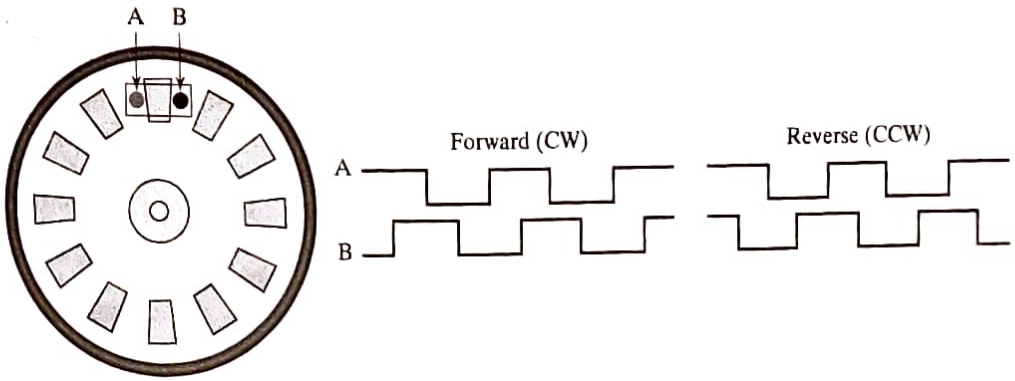


FIGURE 13.61
Quadrature encoding scheme. Left: An encoder wheel with two sensors offset by half the spacing between holes. Right: The output waveforms from each sensor.

The right side of Figure 13.61 shows the wave trains that result from clockwise rotation and counterclockwise rotation. For clockwise rotation, the rising edge of the Channel A precedes the rising edge from Channel B. For counterclockwise, the opposite is true: Channel B's falling edge precedes Channel A's. As a bonus, quadrature encoding increases the resolution of the sensor by a factor of four. For each hole in the encoder disk, a total of four transitions (rising or falling edges) occur for Channels A and B. An optical encoder with quadrature encoding that has 500 holes in the encoder wheel has a resolution of 2000 counts per rotation.

Figure 13.62 shows a typical commercially available optical encoder mounted to a brushed DC motor in a typical assembly (left) and disassembled to show the internal components (right, and below).

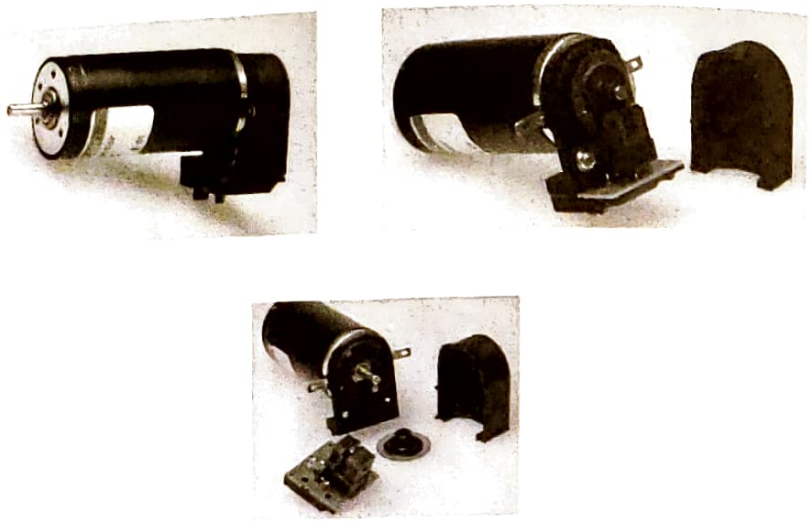


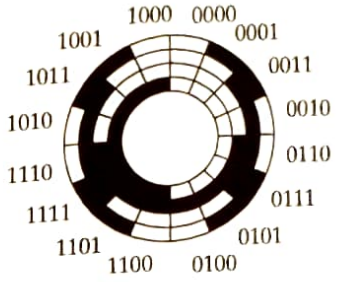
FIGURE 13.62

Top row: A DC motor with encoder (left) and a view of the mounted encoder with the dust cover removed (right). Bottom: DC motor, encoder wheel, sensor assembly and dust cover.

Reflective optical encoders are also available. Reflective encoders bounce the light from an LED off a surface that has alternating reflecting and nonreflecting regions, and use a photo-diode to sense the level of reflected light. Both varieties of encoders are available in rotary and linear configurations.

Optical encoders are also available as absolute position sensors. One method for creating an **absolute optical encoder** is to add more tracks of windows (holes) to the encoder wheel at separate radii that allow for more than simple incrementing or decrementing a count of transitions. For any 22.5° sector of the encoder wheel depicted in Figure 13.63, the pattern resulting from the two white vs. dark regions is unique. If a separate light sensor were used to detect the presence or absence of light shining through each gap, the combination of the sensor outputs is the binary number adjacent to each sector. In this simple example, the absolute angular position of the disk may be determined with a resolution of 1 part in 16, or 22.5°. Higher resolutions are also possible, though space and cost become limiting factors.

For absolute encoder wheels, arranging the slots so that the tracks simply count up or down in binary is an intuitive approach that would work but is rarely used because two or more bits of data would transition simultaneously at several locations. Very nearly, but not quite, simultaneous transitions are about the best we could hope for, and additional logic and error checking would be required to correctly interpret this. To avoid this, the



Position #	Gray Code
0	0000
1	0001
2	0011
3	0010
4	0110
5	0111
6	0101
7	0100
8	1100
9	1101
10	1111
11	1110
12	1010
13	1011
14	1001
15	1000

FIGURE 13.63

A Gray code encoder wheel with 22.5° resolution for absolute position measurements.

code is rearranged so that only one bit changes at a time. This approach is called **Gray code encoding**.

13.5.6.3 Inductive Pickups/Gear Tooth Sensors

Some applications call for measuring the rotation of a rotating gear or shaft, and this can be done with a sensor called an **inductive pickup**, also known as a **gear tooth sensor**. This scheme is popular in the automotive industry to determine crankshaft position and rotational speed (Figure 13.64). Because these sensors tend to be inexpensive and robust, as well as noncontact and aren't optical (they don't have to stay clean to work), they are used in a wide variety of applications across many industries, even though interfacing effectively to inductive pickups can pose some challenges, as we describe below.

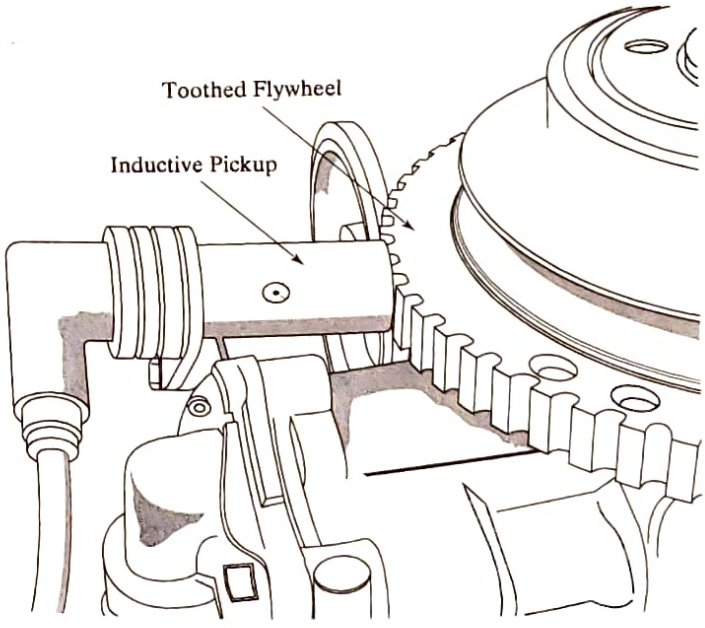


FIGURE 13.64

An inductive pickup used as a crankshaft position sensor in an automobile engine.

Inductive pickups consist of a magnet whose magnetic field is directed through the center of a coil by a flux concentrator (Figure 13.65). When placed in proximity to a ferrous material with an air gap that varies as the ferrous material moves, such as a rotating gear or a pulley with a groove in it, voltage spikes occur across the leads of the coil as shown in Figure 13.65 because of the changes induced in the magnetic field. The magnitude of the output voltage increases as the speed of the ferrous material increases, and also when the air gap is reduced. One of the challenges of interfacing to these sensors is that the output voltages can vary greatly—variation in the 10s of volts is typical—so, care must be taken to ensure that the interface electronics function properly across a wide and often unpredictable range of input voltages, and that the electronics aren't damaged by input voltages well above and below the power supply rails.

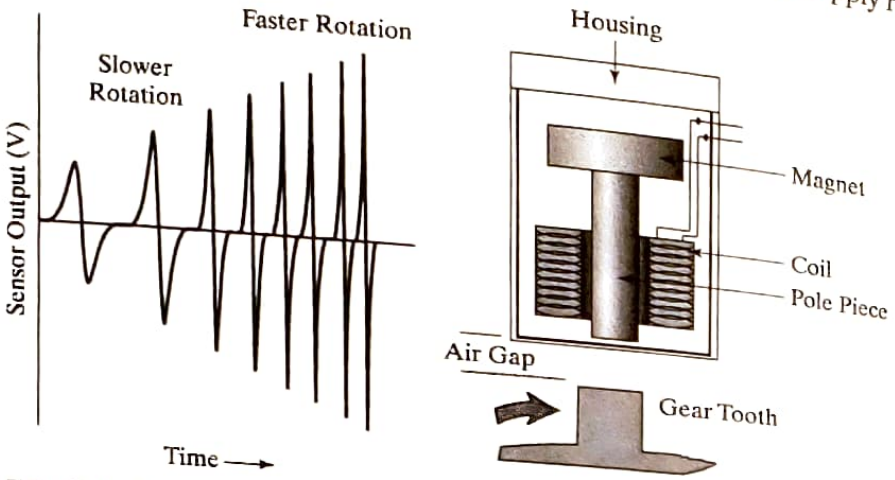


FIGURE 13.65

Construction of an inductive pickup sensor and its characteristic output signal.

13.5.6.4 Reflective Infrared Sensors

We introduced reflective IR sensors in Section 13.5.1.3 and also described their use as proximity detectors in Section 13.5.5. We keep coming back to these sensors because they are compact, inexpensive, easy to integrate, and useful in a very wide variety of applications. In this section, we introduce them as a means of determining the distance to a target.

Figure 13.66 illustrates a typical experimental setup: the emitter and detector are side-by-side in the sensor housing. When an object is relatively close to the emitter (within a centimeter

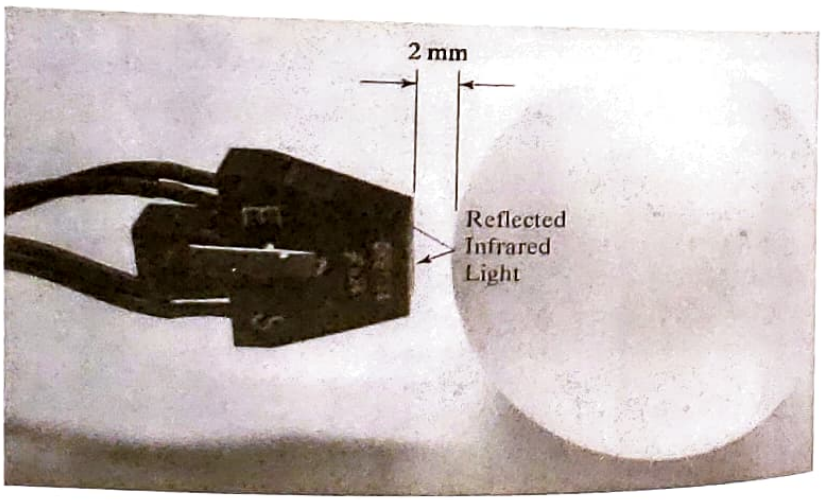


FIGURE 13.66

Using a reflective IR emitter–detector pair sensor to determine distance to a target.

or so), some of the light from the emitter is reflected by the object and sensed by the detector. The amount of light reflected, and hence the amount of collector current that flows, depends on the optical properties of the object and the distance. If the optical properties are known and consistent, then the output depends simply on the distance.

In cases where the system is well characterized and the distances are small, the photo-transistor's collector current will exhibit characteristics similar to that shown in Figure 13.67. which is for the OPTEK® OPB700. When an object (in this case, a specific type of paper, label, or reflective material) is at a distance of 0.1–0.13 in. or less from the sensor, the collector current increases from zero to its maximum value (the plot is normalized). For distances beyond about 0.11 in. (depending on the reflective properties of the material) and up to 1 in. the output drops from the maximum value to about 5% of that value. The response in the region beyond 0.11 in. isn't linear, but the curvature is limited and a linear assumption would be reasonable for small variations in distance.

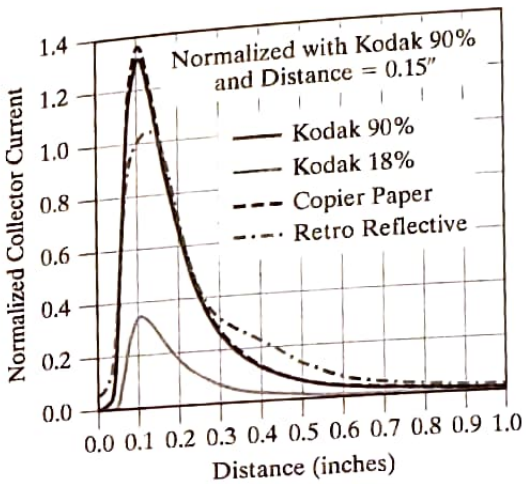


FIGURE 13.67

Normalized collector current vs. distance to various targets for the OPTEK OPB700. (Courtesy of TT Electronics OPTEK Technology.)

13.5.6.5 Capacitive Displacement Sensors

Capacitive displacement sensors provide another means of measuring the distance to an object when that distance is small. In this case, *very* small: capacitive sensors are capable of achieving subnanometer resolution, with a typical maximum range of less than a centimeter. A capacitive displacement sensor is created whenever two conductive objects are brought into close proximity, since this is all that's required to create a capacitor. Recall from Chapter 9 that a simple plate capacitor is made by placing two conductive plates in close proximity. The equation for the resulting capacitance is:

$$C = \frac{\epsilon\epsilon_0 A}{d} \tag{13.17}$$

If we hold the contributions from a dielectric material (the $\epsilon\epsilon_0$ term) and the shared area of the conductors (A) constant, then the capacitance depends only on the distance between the conductors, d. In a simplistic sense, a capacitive displacement sensor consists of one half of a simple plate capacitor, the target makes up the other plate, and the resulting capacitance depends on the spacing between them. Measuring the capacitance is all that's required to determine the distance to the target, and we've discussed several methods to measure capacitance in Section 13.4.6. Subnanometer resolution measurements require an extremely sensitive and low-noise capacitance measurement, which is a difficult challenge. As a result, higher precision measurement requirements are often best met with equipment

purchased from companies focused on capacitive sensing equipment. Lower precision measurements can be made with circuits that we've introduced above. Figure 13.68 shows how a typical capacitive displacement sensor is arranged.

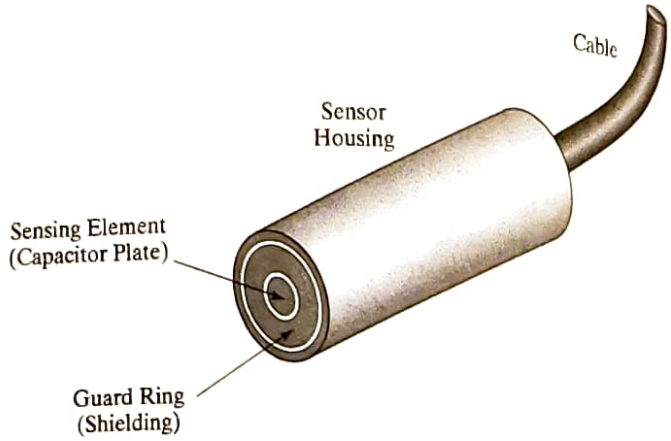


FIGURE 13.68

Capacitive displacement sensor.

In the figure, there are three concentric rings of conductive metal. The sensing element is the inner circle, and the other rings serve to shield the sensing element from capacitive effects with objects other than the intended target. This sensor is a little less than 5 mm in diameter and has a range of 0–10 μm, 15 kHz bandwidth, 0.4 nm resolution, and 0.2% linearity. Those are specifications that are hard to match with anything other than interferometry! In addition to sensing distance to a conductive target, they can be used to sense the distance to nonconductive targets by placing a conductive material on the opposite side. Capacitive sensors are also capable of measuring the thickness of coatings on conductive surfaces, such as adhesives or surface finishes, as well as the thickness of conductive materials. They are, however, sensitive to target size, alignment, and surface irregularities.

13.5.6.6 Ultrasonic Displacement Sensors

We've looked at a few options for measuring short distances with high resolution and accuracy, but what about longer distances? What if an application requires measuring the distance to a target that is greater than a few centimeters away, perhaps even several meters away? This is an application in which ultrasonic range finders excel. These sensors employ the same operating principles as the ultrasonic proximity sensors we discussed above: a high-frequency pulse of sound is emitted by a piezo or electrostatic element, and the time between the emission of the sound and the receipt of its reflection is measured (Figure 13.69). This sensor reports the sound wave's round trip time-of-flight, which is proportional to the distance between the sensor and the object that reflected the sound wave.

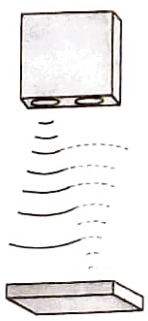
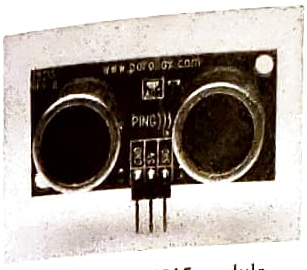


FIGURE 13.69

Ultrasonic waves emitted, reflected, and detected by an ultrasonic range finder.

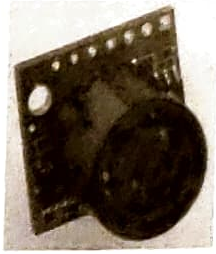
Figure 13.70 shows two ultrasonic range finder modules. The Parallax 28015 (Figure 13.70a) is a fully integrated module that communicates with a host microcontroller via a single digital I/O pin. It has a range of 2 cm to 3 m. Bare ultrasonic transducers, without any interface electronics of software, such as the Maxbotix-UT, are also available (Figure 13.70b). Fully integrated versions using the Maxbotix transducer, with all necessary interface electronics, are also available from the manufacturer. Figure 13.70c shows an example, the Maxbotix LV-EZ1, which has a range of 0–254 in. (6.45 m), and communicates using a digital I/O pin, an analog output pin, or a serial communications link.



(a) Parallax 28015 module



(b) Maxbotix UT transducer



(c) Maxbotix LV-EZ1 module

FIGURE 13.70
Ultrasonic range finders.

13.5.6.7 Flex Sensors

Some applications call for measuring radius of curvature rather than straight-line distance. In some cases, curvature can be related to a quantity you'd like to measure, but isn't the primary measurand, such as arc length or spring deflection. In these cases, a compact, inexpensive sensor called a **flex sensor** or **bend sensor** may be a good choice (Figure 13.71). Flex sensors are made by printing a thin coating of a material that forms a resistive element onto a flexible substrate such as mylar or polyimide. The entire sensor can be as little as 0.004 in. thick. When the sensor is bent or flexed, microscopic cracks form in the printed sensing material, and the bulk resistance increases as the radius of curvature decreases. That is, the more they are flexed, the higher their resistance. A typical flex sensor has a nominal unflexed resistance of around 10 kΩ, and the resistance may rise well to hundreds of kilohm at high curvatures.

Flex sensors have strong temperature and humidity dependencies, so they are not typically used for high-precision measurements or applications that require stability. They are best for measurements that involve relatively fast changes and for those where relative changes in radius of curvature are more important than quantifying the actual radius. Examples of typical applications include interface devices that measure body motion for computer games and virtual reality simulations, automotive horn buttons, and sensing if a passenger is seated in a car for air bag activation/deactivation.

13.5.7 Acceleration Sensors

As of 2008 (when this chapter was written), measurements of acceleration could not be performed directly with a material whose properties change solely as a function of an applied acceleration. Instead, acceleration is inferred by measuring another quantity that is influenced by acceleration, such as the motion of a mass attached to a spring element. Earlier in this chapter, we introduced the most common way this is sensed in commercially available